

Kabir Muzumdar

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Education

University of Texas at Austin

August 2024 – May 2028

Bachelor of Science, Civil Engineering (CE) — Certificate, Computational Engineering & Science (CSE)

- GPA: 3.7/4.0
- **Relevant Coursework:** Mechanics of Solids, Fluid Mechanics, Structural Analysis, Ocean/Offshore Engineering.

Research Experience

Optimization of Damping Performance for an Origami-Inspired Metasurface

Austin, TX

Jan 2025 – Present

Undergraduate Research Assistant: Mentor: Othman Oudghiri-Idrissi, UT Austin

- Developed parametric and finite-element models (FEM) of deployable structures to advance research in structural dynamics, wave propagation, and aerospace structural mechanics.
- Designed compliant hinge systems and visco-elastic damping concepts to improve deployability, vibration performance, and structural efficiency.

Dynamically Stable Large Space Structures via Architected Metamaterials

Austin, TX

Mar 2026 – Present

Undergraduate Research Assistant: Mentor: Othman Oudghiri-Idrissi, UT Austin & NASA Ames Research Center Collaboration

- Designed deployable voxel architectures for a NASA Ames-led In-space Assembly & Manufacturing (ISAM) collaboration with UT Austin and the University of Michigan.
- Developed bidirectional hinge and interlocking mechanisms to improve structural retention and enable robotic assembly of modular space structures for NASA's Starshade proposal to support exploration of exoplanets and deep space.

Design of Thruster for High Efficiency & Analyzed Through Advanced Computing Methods

Austin, TX

July 2026 – Present

Undergraduate Research Assistant: Mentor: Spyridon Kinnas, UT Austin — Sponsored by UT Applied Research Laboratories, UT Austin Office of the Vice President for Research

- National Security Research SEED funding from ARL:UT and OVPR at UT Austin.
- Using software such as PROPCAV, Tecplot, and Fluent for Computational Fluid Dynamics (CFD) analysis of high-speed ducted thrusters for military applications, with an emphasis on cavitation and fluid-structure interaction (FSI).

Posters & Presentations

Cockrell Distinguished Alumni Undergraduate Research Fellowship —
Dynamic Analysis of 3D Extruded Geometries of Miura-ori Solar Arrays with Emphasis on Flexural Vibration Modes

May 2025

Team: **Kabir Muzumdar**, Othman Oudghiri-Idrissi

Submitted to the Engineering Academy of Distinguished Alumni & Dr. Manish Kumar at the Fariborz Maseeh Department of Civil, Arch., Envir. Engineering, UT Austin.

Projects

Parametric Origami (Miura-ori) Metasurface

- Developed a parametric model of a Miura-ori metasurface with parameters for tunability and material definition.
- Performed eigenfrequency analysis on the metasurface with a focus on geometric tunability and deployable design.
- Tools & Methods Used: COMSOL Multiphysics, MATLAB, Eigenfrequency Analysis.

Parametrically-defined Voxel Framing and Structure

- Designed a parametric lattice-structure "voxel" model for the Dynamically Stable Large Space Structures project.
- Compared multiple iterations of the design with differing advantages in the context of robotic in-space assembly and manufacturing (ISAM).
- Tools & Methods Used: SolidWorks, Parametric Modelling, Equation-based Modelling

Floquet-Bloch Analysis of Origami-based In-space Structures

- Analyzed a reshaped periodic cell of an origami structure within the Brillouin Zone of its propagation behavior.
- Used numerical methods like the Irreducible Brillouin Zone (IBZ) to make models computationally efficient and allow for understanding the behavior through the structure using periodic boundary conditions and wave-vector sweeps.

Drone-based Evaluation of Damage on Structures & Roadways

- Evaluated roadway-image datasets for object-detection training, with emphasis on representative crack classes, drone-perspective imagery, annotation quality, and deployment-relevant variation.
- Developed recommendations for positive/negative image balance, hard-negative selection, train/validation/test splitting, and augmentation to reduce false detections in roadway damage classification.
- Supported Roboflow/YOLO dataset preparation and model-evaluation workflow for a collaborative roadway-inspection project.
- Tools & Methods Used: Roboflow, YOLO, Dataset Engineering, Object Detection, Data Augmentation.

Leadership, Extracurricular Activities, & Service

Structural & Design Lead | *American Society of Civil Engineers* | *Canoe*

- Mentored Civil Engineering students in SolidWorks basic part design, assembly mating fundamentals, and FLOW to analyze designs using CFD.
- Developed project documentation for Canoe design with Structural, QA/QC, and Logistics leads for proposal submission.

Ambassador | *Autodesk* | *Design & Make Program*

- Represented Autodesk on UT Austin campus, spreading awareness of company events and opportunities.
- Hosted workshops, gave presentations, and collaborated with campus organizations involved in engineering, design, architecture, construction, and manufacturing on behalf of Autodesk.

Director of Civil Engineering | *UT Austin Sustainable Building Initiative*

- Led teams of 30+ engineers to work on projects within mechanical and civil/architectural engineering for club projects.
- Mentored students in designing, drafting, and using tools like Revit, SketchUp, and AutoCAD.
- Focused on representing client vision to professional engineers through drafts and drawings.

Technical Skills

Software & Programming: COMSOL Multiphysics, SOLIDWORKS, MATLAB, Python, Java, AutoCAD, Fusion, Revit, Rhino, Git, GitHub, LaTeX.

Technical Methods: Finite Element Analysis (FEA), Computational Mechanics, Structural Analysis, Computational Fluid Dynamics (CFD), CAD & Parametric Modeling, Mechanical Design and Fabrication, Laser Vibrometry.

Additional

Research Interests: Computational Mechanics, Structural Dynamics, Fluid-Structure Interactions, Mechanical Metamaterials, Engineering Simulation.

Honors & Awards: Cockrell Distinguished Alumni Scholarship; Maseeh Department Undergraduate Research Fellowship; Halff Associates, Inc. Endowed Scholarship; Linford Family Endowed Scholarship.

Languages: English (Native or Bilingual Proficiency); Spanish (Limited Working Proficiency); Marathi (Professional Working Proficiency); Hindi (Limited Working Proficiency).